



COMSATS University, Islamabad Attock Campus

Program: BS (CE) – V

Fall 2024

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Course: Principle of Communication Systems

COMPLEX ENGINEERING PROBLEM

Assignment = 03 & 04 (CLO2-PLO2-C2) (CLO3-PLO3-C5)

Questions: Design, Analysis and Investigation:

In the **Principles of Communication Systems** course, complex engineering problems typically involve designing, analyzing, or optimizing communication systems that require a deep understanding of the underlying principles. Here's an example of a complex engineering problem within this context:

Problem: **Designing a Reliable Digital Communication System for a Noisy Channel**

Scenario:

You are tasked with designing a digital communication system that reliably transmits data over a wireless channel. The channel introduces noise, and there are bandwidth and power constraints. Your goal is to minimize the bit error rate (BER) while maximizing the data rate within the given constraints.

Specifications:

- **Channel Type:** AWGN (Additive White Gaussian Noise)
- **Bandwidth:** 10 MHz
- **Transmitter Power:** 5 Watts
- **Required BER:** Less than 10^{-6}
- **Modulation Scheme:** To be chosen (options include BPSK, QPSK, 16-QAM, etc.)
- **Data Rate:** Maximize, while maintaining the required BER.
- **Coding:** Error correction codes are allowed (e.g., Convolutional Codes)
- **Receiver Sensitivity:** To be optimized.

Tasks:

1. **Modulation Scheme Selection:** Choose a modulation scheme that balances data rate and BER performance in the presence of noise. Justify your choice.
2. **Error Control Coding:** Incorporate an error correction coding scheme to reduce the BER below 10^{-6} in a noisy environment. Determine the coding gain and how it affects system performance.

3. **Power and Bandwidth Optimization:** Analyze how power and bandwidth affect the performance of the chosen modulation scheme and error correction code. Make adjustments to optimize the system.
4. **BER Analysis:** Derive the theoretical BER for your chosen modulation and coding scheme over the AWGN channel. Use this to evaluate whether the design meets the BER requirement.
5. **Simulation:** Implement a simulation of the system (using MATLAB, Python, or another tool) to validate the performance of your design under various noise conditions.
6. **Trade-off Analysis:** Discuss the trade-offs between increasing the data rate and minimizing the BER, considering factors like modulation complexity, power usage, and bandwidth constraints.

Deliverables:

- Mathematical derivation of BER for the chosen modulation scheme.
- Simulation results showing the performance of the system (data rate, BER) under different noise conditions.
- A report explaining the design choices, trade-offs, and how the system meets the required specifications.

Key Concepts Involved:

- **Signal-to-Noise Ratio (SNR):** Analyze how SNR affects BER in digital modulation schemes.
- **Modulation Techniques:** Binary Phase-Shift Keying (BPSK), Quadrature Amplitude Modulation (QAM), etc.
- **Error Correction Codes:** Incorporating Forward Error Correction (FEC) to improve system reliability.
- **Shannon's Capacity Theorem:** Use this to analyze the theoretical limits of data transmission over a noisy channel.
- **Nyquist Criterion:** Evaluate the bandwidth efficiency of different modulation schemes.

This type of problem combines theoretical analysis, system design, and simulation, all while considering real-world constraints like noise, power, and bandwidth—making it a complex and challenging engineering task within the course.

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Solution:

Problem: Designing a Digital Communication System for a Noisy Channel

Scenario:

You are tasked with designing a digital communication system that transmits data over a noisy channel. The system should:

1. **Maximize data rate** while ensuring reliable transmission.

2. **Minimize bit error rate (BER)** despite the presence of additive white Gaussian noise (AWGN).
3. **Utilize a bandwidth of 10 MHz and signal-to-noise ratio (SNR) of 10 dB.**
4. Choose an appropriate **modulation scheme** and **error-correcting code**.

Solution:

1. Signal-to-Noise Ratio (SNR) and Channel Capacity:

Using **Shannon's Capacity Theorem**, the maximum data rate for a given bandwidth and SNR can be computed as:

$$C = B \log_2(1 + \text{SNR})$$

Where:

- CCC is the channel capacity (maximum data rate in bits per second).
- BBB is the channel bandwidth (10 MHz in this case).
- S/N is the SNR, converted from dB to linear scale:

$$\text{SNR} = 10^{\frac{10}{10}} = 10$$

Thus, the channel capacity is:

$$C = 10 \times 10^6 \log_2(1 + 10) = 10 \times 10^6 \log_2(11) \approx 10 \times 10^6 \times 3.4594 \approx 34.59 \text{ Mbps}$$

So, the theoretical maximum data rate is approximately **34.59 Mbps**.

2. Modulation Scheme:

To achieve reliable communication at this data rate, we need to select a modulation scheme that balances data rate and error performance. Common modulation schemes include:

- **BPSK (Binary Phase Shift Keying):** Offers robust performance in noisy channels but low spectral efficiency.
- **QPSK (Quadrature Phase Shift Keying):** Doubles the data rate compared to BPSK by encoding 2 bits per symbol.
- **16-QAM (Quadrature Amplitude Modulation):** Further increases data rate but is more susceptible to noise.

Given the channel conditions (SNR = 10 dB), **QPSK** is a suitable choice because it provides a good balance between data rate and robustness in noisy environments.

With **QPSK**, the system can transmit **2 bits per symbol**. The symbol rate for QPSK is:

$$\text{Symbol Rate} = \frac{\text{Data Rate}}{\text{Bits per Symbol}} = \frac{34.59 \times 10^6}{2} \approx 17.3 \text{ Msymbols/sec}$$

$\approx 17.3 \text{ Msymbols/sec}$ Symbol Rate=Bits per Symbol Data Rate= $234.59 \times 106 \approx 17.3 \text{ Msymbols/sec}$

Thus, **QPSK** will allow us to achieve the desired data rate within the given bandwidth.

3. Error-Correcting Code:

To further reduce the bit error rate (BER), an **error-correcting code** should be applied. A common choice is **Convolutional Coding** combined with **Viterbi Decoding**, which can offer significant BER improvements without too much additional complexity.

By selecting a **rate 1/2 convolutional code**, the data rate will be halved, but the system will be much more resilient to noise. Therefore, the effective data rate becomes:

Effective Data Rate= $34.592 \approx 17.3 \text{ Mbps}$
 $\text{Effective Data Rate} = \frac{34.59}{2} \approx 17.3 \text{ Mbps}$

This rate is still sufficient given the system constraints and will significantly improve error performance.

4. Bit Error Rate (BER) Performance:

For QPSK over an AWGN channel, the approximate BER is given by:

$\text{BER} \approx Q(\sqrt{2 \frac{E_b}{N_0}})$
 $\text{BER} \approx Q(2\sqrt{N_0 E_b})$

Where:

- E_b/N_0 is the energy per bit to noise power spectral density ratio.

With an SNR of 10 dB, the BER can be calculated using standard Q-functions or approximated via simulation (e.g., MATLAB). In practical terms, with QPSK and convolutional coding, the system can achieve a **BER of less than 10^{-6}** , which is considered highly reliable.

Conclusion:

The designed system uses **QPSK modulation, rate 1/2 convolutional coding**, and operates at an effective data rate of **17.3 Mbps**. This configuration ensures reliable communication over the noisy channel with a **low BER** and **maximized data rate** within the 10 MHz bandwidth constraint.

This type of problem demonstrates key principles in communication systems, including channel capacity, modulation, coding, and error performance analysis.